

Calculating the Volume of Cylinders

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A **cylinder** is a solid figure with two congruent circular bases connected by a curved surface.

Volume of a Cylinder

The **volume** of a cylinder is the amount of space inside it.

The **base** of a cylinder is a circle with area

$$A = \pi r^2$$

Height

The **height** is the perpendicular distance between the two circular bases.

$$\text{Height} = h$$

Volume Formula

The formula for the **volume of a cylinder** is

$$V = \pi r^2 h$$

where r is the radius and h is the height.

When Diameter Is Given

If the **diameter** is given, first find the radius using

$$r = \frac{d}{2}$$

Then substitute the radius and height into

$$V = \pi r^2 h$$

Examples

Example 1

Find the volume of a cylinder with radius 4 cm and height 10 cm.

$$V = \pi r^2 h$$

$$V = \pi(4)^2(10) = 160\pi$$

Therefore, $V = 160\pi \text{ cm}^3$.

Example 2

Find the volume of a cylinder with diameter 14 cm and height 15 cm. Use $\pi = \frac{22}{7}$.

$$r = \frac{14}{2} = 7 \text{ cm}$$

$$V = \frac{22}{7}(7)^2(15)$$

$$V = 2310 \text{ cm}^3$$

Example 3

A cylindrical container has a volume of 1 256 cm³ and a radius of 10 cm. Find its height.

$$V = \pi r^2 h$$

$$1256 = \pi(10)^2 h$$

$$h = \frac{1256}{100\pi} \approx 4 \text{ cm}$$

Practice Exercises

Practice 1

Find the volume of a cylinder with radius 4 cm and height 10 cm.

Practice 2

Find the volume of a cylinder with radius 6 cm and height 8 cm.

Practice 3

Find the volume of a cylinder with radius 7 m and height 5 m.

Practice 4

Find the volume of a cylinder with diameter 14 cm and height 15 cm. Use $\pi = \frac{22}{7}$.

Practice 5

Find the volume of a cylinder with diameter 18 cm and height 9 cm.

Practice 6

A water tank is shaped like a cylinder with radius 2.5 m and height 4 m.

Find its volume.

Practice 7

A candle is cylindrical with radius 3 cm and height 18 cm.

Find its volume.

Practice 8

A cylindrical can has diameter 12 cm and height 20 cm.

Find its volume.

Practice 9

A pipe has an inside radius of 15 cm and height 2 m.

Find its volume.

Practice 10

A cylindrical container has a volume of $1\,256\text{ cm}^3$ and a radius of 10 cm .

Find its height.

Activities

Activity 1

Find the volume of a cylinder with radius 5 cm and height 9 cm.

Activity 2

Find the volume of a cylinder with radius 8 cm and height 7 cm.

Activity 3

Find the volume of a cylinder with radius 9 m and height 4 m.

Activity 4

Find the volume of a cylinder with diameter 16 cm and height 12 cm. Use $\pi = \frac{22}{7}$.

Activity 5

Find the volume of a cylinder with diameter 20 cm and height 14 cm.

Activity 6

A cylindrical storage drum has radius 3 m and height 5 m.

Find its volume.

Activity 7

A roll of paper is cylindrical with radius 4 cm and height 15 cm.

Find its volume.

Activity 8

A cylindrical bucket has diameter 18 cm and height 16 cm.

Find its volume.

Activity 9

A cylindrical pillar has radius 20 cm and height 3 m.

Find its volume.

Activity 10

A cylindrical container has a volume of 2512 cm^3 and a radius of 10 cm .

Find its height.

Thank you!